

The energy transition

The Earth is getting warmer.

In 2023, according to the [United Nations World Meteorological Organization](#), the global annual average temperature was nearly 1.5 degrees Celsius **above pre-industrial levels** (more precisely, 1.45 ± 0.12 °C).

To achieve improvement, our main tool is **the energy transition**, i.e., the shift from an energy mix based on fossil fuels to one that produces very limited, if not zero, carbon emissions, based on renewable energy sources. A huge contribution to [decarbonization](#) comes from the electrification of consumption, replacing fossil fuel-generated electricity with energy generated from [renewable sources](#), which also makes other sectors like transport cleaner; the digitalization of grids also contributes by improving energy efficiency.

1.45°

The temperature increase in 2023 compared to pre-industrial levels

1.5°

The rise in temperature by 2030, if no steps are taken

97%

The percentage of scientists who attribute global warming to human activities

1: ELECTRIFICATION

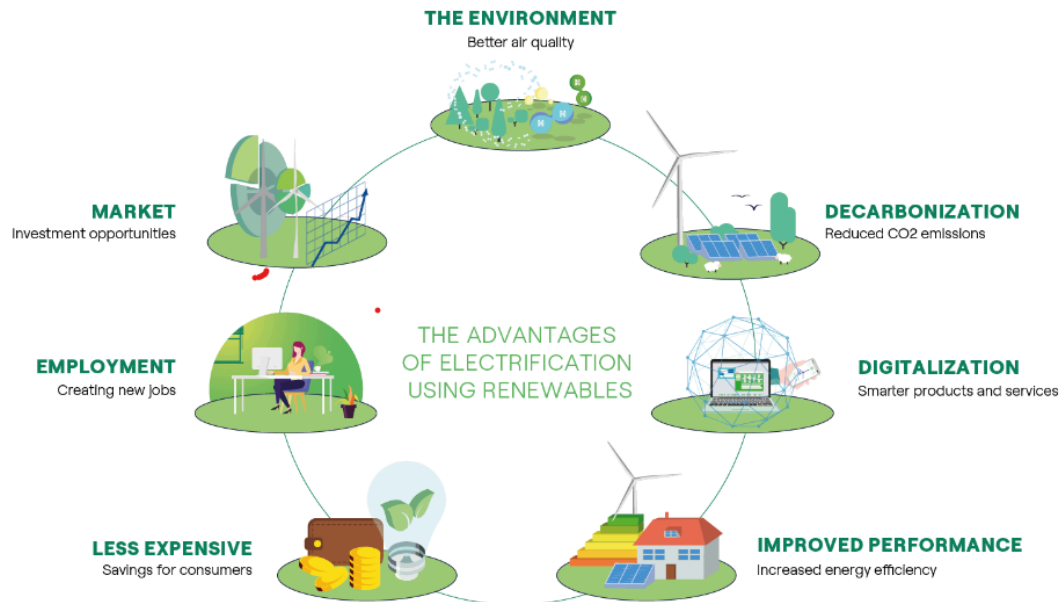
Renewables spearheading electrification

“Electrification” means gradually transitioning to using electricity generated by renewables for **public and private services and activities** which have until now been powered mostly by fossil fuels. Imagine a cleaner, more sustainable and efficient world with everything - from our homes to transport and industry - powered just by **decarbonized electricity**.

[Electrifying final consumption](#) using energy from [renewables](#) is the most effective way to bring about the [energy transition](#). It is only by [decarbonizing](#) electricity generation that we will be able to halt [climate change](#), which is currently threatening the planet.

The direct electrification of final consumption using renewable energy can also be combined with the **indirect electrification** of so-called '**hard-to-abate**' industrial sectors – shipping, aviation, mining and heavy industry– using green hydrogen.

The advantages of electrification using renewables



2: DIGITALIZATION



The energy transition currently under way embraces all aspects of the electricity system. One of the most disruptive of these, together with the decarbonization of the electricity generation mix, is **digitalization**, which is transforming the processes by which energy is produced, distributed and consumed.

Smart grids that are increasingly flexible and decentralized

The most evident effect of digitalization concerns the **grids** that transport and distribute the electricity that is produced. On this front the basic elements are **electronic meters** that enable **smart grids**, with which it is possible to manage and balance the electricity system efficiently. That is a solution that is particularly important for intermittent renewable sources, such as wind or solar power, which can thereby be fully integrated with the grid. This has an even greater value in a scenario in which the network becomes increasingly flexible and decentralized. The old top-down structure, in which energy flows in a single direction from producer to consumer, is now shifting towards a model of distributed generation in which there is a growing number of small producers and consumers capable of feeding electricity into the network. **Flexibility** thus becomes a key element that enables grids to manage flows and to establish a dialogue in real time between operators and consumers, who are increasingly becoming fully-fledged actors on the grid. The challenge of the digitalization of the energy sector therefore mainly concerns the **grid operators**. Estimates indicate that 17% of investment in the networks in future will focus on innovations in terms of **transmission** (TSO, Transmission System Operator) and, above all, **distribution** (DSO, Distribution System Operator) from renewable sources.

The other significant evolution of smart grids is their transformation into **resilient, participatory and sustainable platforms** to facilitate new services and generate value to the benefit of the entire ecosystem. The infrastructure thus becomes the essential basis for economic and social activities. Electricity distribution networks that are accessible, flexible and inclusive play a central role in enabling society in general, and stakeholders in particular, to achieve their goals. Moreover, the **intelligent management** of the network boosts efficiency and reduces waste, resulting in further benefits for the environment.

Finally, the new model for electric energy – with the electrification of consumption, i.e. replacing technologies that use fossil fuels with technologies that use electricity from renewable sources – opens up opportunities in terms of the market and business models, models that involve old and new players and that require transparency and the real time tracking of energy flows. Grid operators, then, position themselves as neutral facilitators of this dynamic market and, above all, as active partners in energy storage and the management of energy loads, in addition to monitoring electricity rates.

A new role for the consumer

As we have seen, at the other end of the energy system, that of the final consumer, digitalization is also fostering the energy transition. An example of the tangible benefits that digitalization offers customers is represented by the innovative interfaces through which new smart meters provide information about consumption and production practically in real time, enabling new services such as **demand response** and **home automation**, remotely operated intelligent solutions that manage security systems, electrical appliances and temperature regulation.

The role of customers is changing. From passive and largely unaware users they are becoming **active and discerning protagonists in the electricity system**, increasing their own energy awareness. This is even more evident in the case of “prosumers”, i.e. customers who are both producers and consumers of energy. Thanks to digitalization they can contribute to the electricity generation mix and help reduce emissions.

The energy transition is a phenomenon that goes beyond the simple generation of clean electricity. Through digitalization it involves everyone, producers and consumers.

3: DECARBONIZATION

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